1. LED blinking patterns

Blinks 4 LEDs in different patterns. Uses **GPIO.BOARD** (physical pin numbers).

```
LED 1 -> Pin 7
LED 2 -> Pin 11
LED 3 -> Pin 13
LED 4 -> Pin 15
(each LED through a resistor to GND)
```

Practical 2 — Telegram control of GPIO

Module pin / item	Raspberry Pi connection
Green LED positive	BCM GPIO2 / physical pin 3
Yellow LED positive	BCM GPIO3 / physical pin 5
Red LED positive	BCM GPIO4 / physical pin 7
Blue LED positive	BCM GPIO17 / physical pin 11
All LED negatives	GND (source manual says physical pin 9)

Practical 3 — Display current time on TM1637 4-digit display

Module pin / item	Raspberry Pi connection
TM1637 GND	Physical pin 14 — GND (as source manual)
TM1637 VCC	Physical pin 4 — 5V
TM1637 DIO	Physical pin 18 — BCM GPIO24
TM1637 CLK	Physical pin 16 — BCM GPIO23

4. GPS module

Prints GPS data over serial (**/dev/serial0**, 9600 baud). TX and RX are crossed.

```
GPS GND -> Pin 6
GPS VCC -> Pin 4  (5V)
GPS RXD -> Pin 8  (TXD)
GPS TXD -> Pin 10 (RXD)
```

5. Oscilloscope (ADS1115)

The ADC reads an analog signal and draws a live graph. Enable I2C. Uses **matplotlib** and **drawnow**.

```
VDD -> Pin 17 (3.3V)
GND -> Pin 9
SCL -> Pin 5  (GPIO 3)
SDA -> Pin 3  (GPIO 2)
```

6. 16x2 I2C LCD

Shows your own text on the LCD. Check the address with **i2cdetect -y 1** (0x27 or 0x3F).

```
GND -> Pin 6
VCC -> Pin 4 (5V)
SDA -> Pin 3 (GPIO 2)
SCL -> Pin 5 (GPIO 3)
```

7. RFID read/write/erase

Reads, writes and erases tag data. Enable SPI. This module needs **3.3V, not 5V**.

```
VCC  -> Pin 1  (3.3V)
GND  -> Pin 6
SDA  -> Pin 24 (CS0)
SCK  -> Pin 23
MOSI -> Pin 19
MISO -> Pin 21
```

Practical 8 — Fingerprint sensor interfacing

Module pin / item	Raspberry Pi connection
Fingerprint VCC (red)	Physical pin 4 — 5V (source manual)
Fingerprint GND (black)	Physical pin 6 or 9/14 — GND; follow the specific manual/module
Sensor TXD	Physical pin 10 — BCM GPIO15 / UART RXD
Sensor RXD	Physical pin 8 — BCM GPIO14 / UART TXD

11. Fingerprint + LCD

Run **enroll.py** first, then **main_verify.py**. The LCD shows MATCHED or UNMATCHED. Enable both I2C and serial.

```
LCD:
GND -> Pin 6
VCC -> Pin 2 (5V)
SDA -> Pin 3
SCL -> Pin 5

Fingerprint sensor:
Red (VCC)       -> Pin 4
Black (GND)     -> Pin 9 or 14
Yellow/Green TX -> Pin 10
White/Blue RX   -> Pin 8
```

12. Fingerprint door lock (relay)

A matched fingerprint turns the relay on for 5 seconds. The relay is **active-LOW**, so LOW turns it ON.

```
Relay:
VCC -> Pin 2
GND -> Pin 9
IN  -> Pin 11 (GPIO 17)

Fingerprint sensor:
VCC -> Pin 2 (5V)
GND -> Pin 6
TX  -> Pin 10
RX  -> Pin 8
```

13. MQTT LED control

The MQTT Dashboard phone app publishes "on" or "off" to topic **led**. Broker: **broker.emqx.io**, port 1883.

```
LED (VCC) -> Pin 12 (GPIO 18)
GND       -> Pin 6
```

14. Flask web relay

A web page with ON and OFF buttons controls the relay. Open **Pi's IP:5000**.

```
VCC  -> Pin 2
GND  -> Pin 6
IN 1 -> Pin 12 (GPIO 18)
```

Your PDF's table says GPIO 18, but the code uses **RELAY = 17**. Check which pin you actually wired.

17. Node-RED button + LED

A physical button and a web dashboard both toggle the same LED. Start it with **node-red-start**.

```
Your PDF gives no pin numbers.
Push button -> any GPIO you choose
LED         -> any GPIO you choose
(set the same pins in the GPIO in / GPIO out nodes)
```

Practical 19 — Embedded RFID/fingerprint access control

Module pin / item	Raspberry Pi connection
Relay VCC / GND / IN	5V physical pin 2 / GND physical pin 9 / BCM GPIO23 physical pin 16
LCD GND / VCC	Physical pin 6 GND / physical pin 4 5V
LCD SDA / SCL	Physical pin 3 GPIO2 SDA / physical pin 5 GPIO3 SCL
Fingerprint VCC / GND	5V / GND
Fingerprint TX / RX	GPIO15 RXD physical pin 10 / GPIO14 TXD physical pin 8
RC522 3.3V / GND	Physical pin 1 / physical pin 6
RC522 RST	Physical pin 22 — BCM GPIO25
RC522 MISO / MOSI / SCK / SDA-SS	Physical pins 21 / 19 / 23 / 24 — BCM GPIO9 / GPIO10 / GPIO11 / GPIO8
LED	BCM GPIO21 per source code; physical pin not specified in source table